

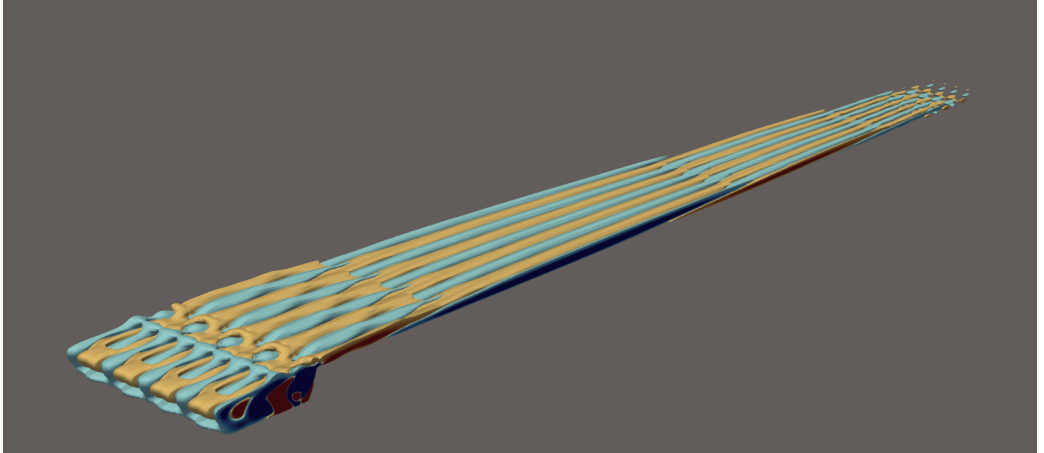
IMPERIAL

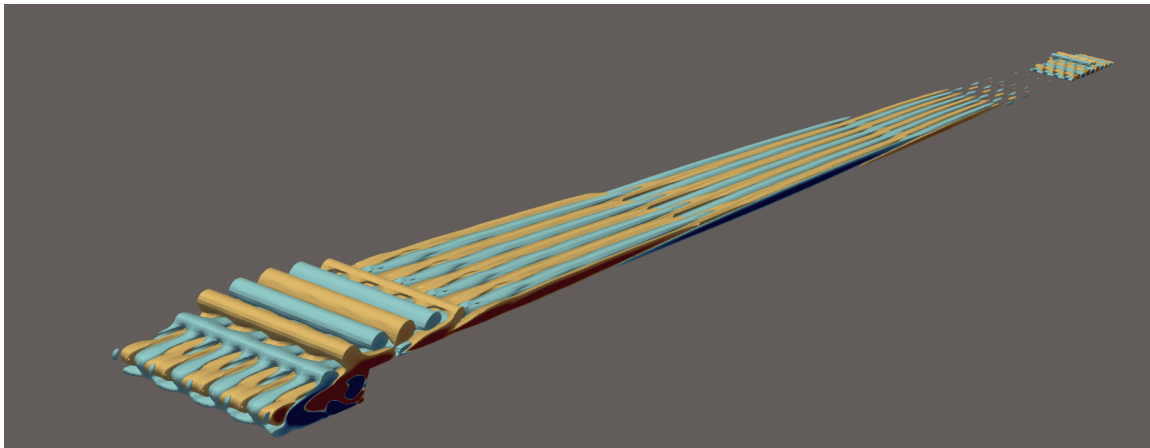
Re analysis

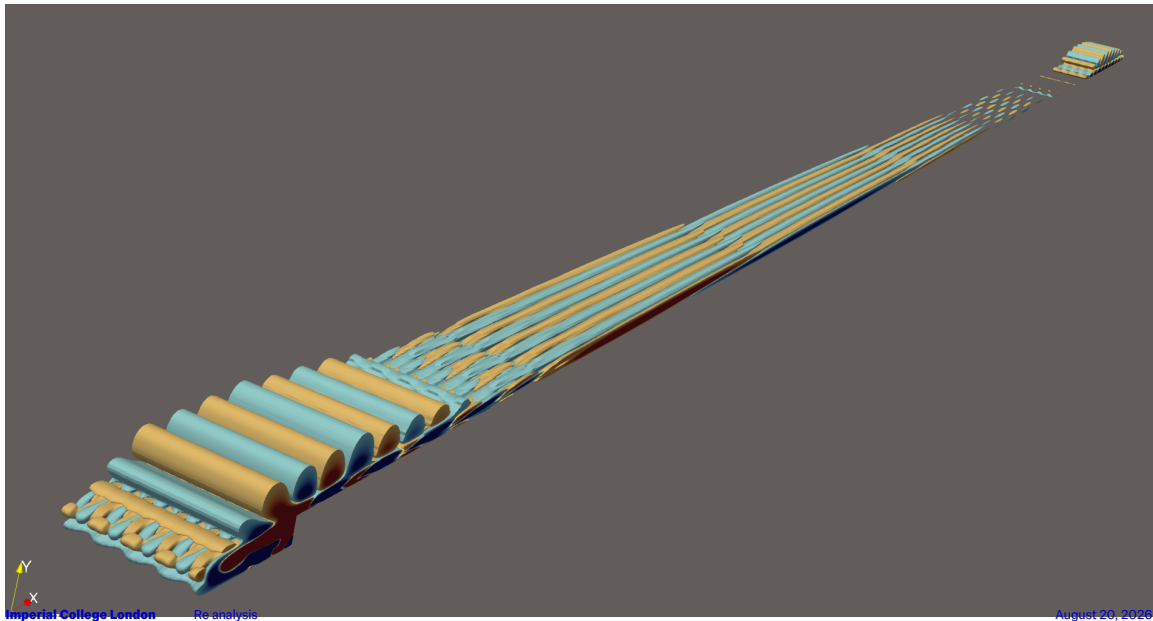
Víctor Ballester
August 20, 2026

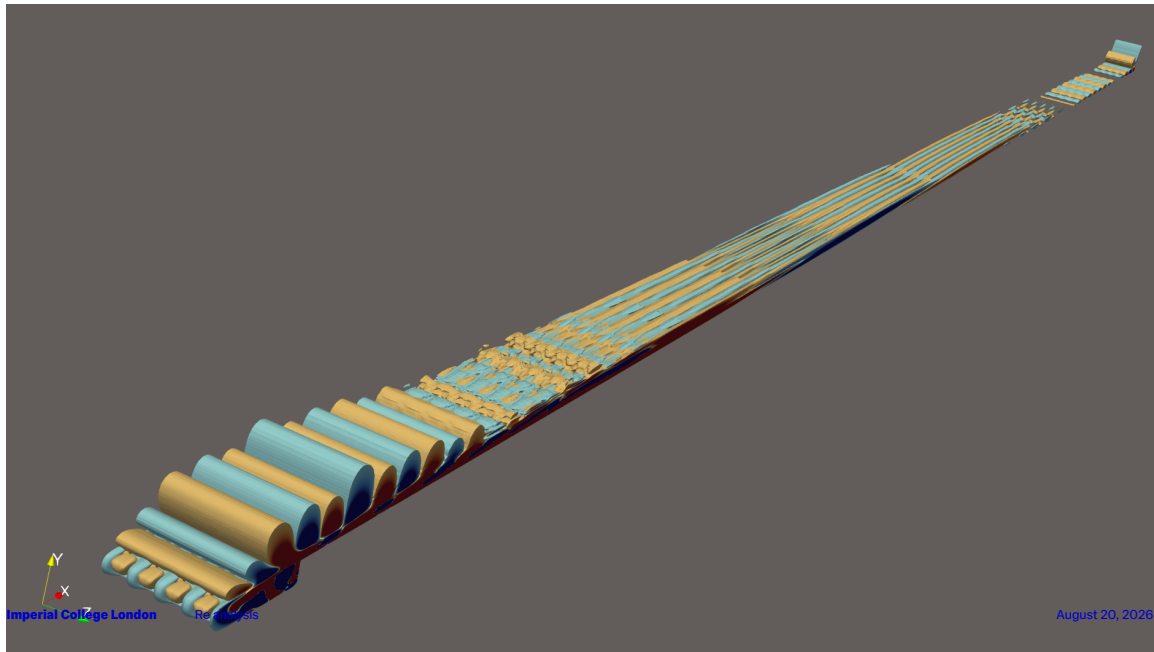
Transition starting from 3d mode

I start the simulation with the baseflow and 3d mode. And the small noise of the mesh and 3d noise is enough to trigger the transition.



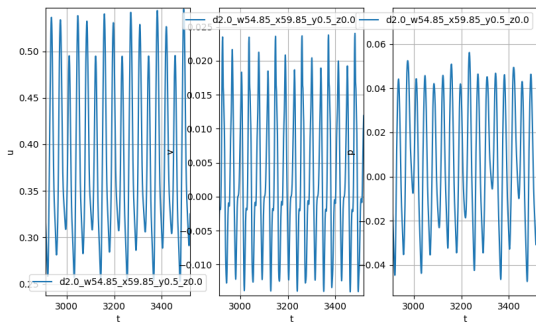




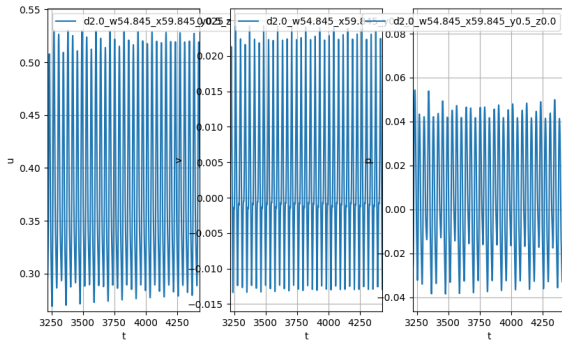


secondary bifurcation in 2d dynamics

- When I run floquet analysis (with respect to the non-linear stable P.O.) I find two modes. The first one is the one that evolves to the P.O. that I linearized about (the nonlinear period is slightly different from the linear one). The second mode is a new mode.
- freq mode 1: $\omega_1 = 0.174$ (frequency of the P.O. I linearize around is $\omega = 0.169$)
- freq mode 2: $\omega_2 = 0.216$
- I don't know whether the difference $\omega_2 - \omega_1$ is related to $\simeq 1/3\omega_1$, which is what I observed on the time periodic signals around the bifurcation point.



Time series destabilising when starting around the periodic orbit (showing the $3T$ period)



Time series stabilising when starting around the periodic orbit (showing the $3T$ period stabilizing to the $1T$ period)